Jai S Ramani

AI & ML Engineer | Data Analyst

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PROFILE SUMMARY

Analytical and results-driven undergraduate with hands-on expertise in data analytics, business intelligence, and database querying. Proficient in transforming raw, multi-source datasets into actionable insights using SQL, Python (Pandas, NumPy), Excel, Power BI, and Tableau. Experienced in building automated data extraction pipelines, time-series analysis, statistical KPI tracking, and interactive dashboards, complemented by strong foundations in applied machine learning and predictive modeling. Seeking to drive data-informed business decisions through rigorous exploratory analysis and quantitative problem-solving.

SKILLS

Databases: SQL (SQLite, PostgreSQL), MongoDB Atlas, Supabase
Frameworks & Libraries: Pandas, NumPy, OpenCV, React, Node.js, Express, Flask, Django REST
Programming Languages: Python, SQL, Java, JavaScript
Tools & Platforms: Excel, Power BI, Tableau, Docker, Git, GitHub, Pytest, Playwright
Soft Skills: Communication, Teamwork, Problem Solving, Adaptability

PROJECTS

PricePilot — Price Intelligence & Arbitrage Platform · *Python, Flask, SQLite, Pandas, Docker* *GitHub*

- Architected an end-to-end price intelligence & arbitrage platform tracking time-series product data across **5 Amazon regions** and benchmarking prices against **18 retail marketplaces** (Flipkart, Croma, Blinkit, Walmart).
- Engineered a **6-engine analytics suite** leveraging 30-day moving averages, price volatility modeling, and promotional bank offers to generate Deal Quality Scores, price-drop forecasts, and real-time BUY / WAIT signals.
- Built a high-performance **Flask backend** with a multithreaded daemon scheduler for autonomous polling, SQLite/CSV pipelines, and real-time alert dispatch via HTML Email, Discord Webhooks, and Telegram Bots.
- Developed an interactive dashboard utilizing **Chart.js** and **GSAP** for responsive 30-day time-series analytics, inline sparklines, split-basket cost optimization, and multi-currency normalization across **7 fiat currencies**.
- Containerized with **Docker & Docker Compose** and maintained a 100% pass rate across **59 automated unit and integration tests** with **pytest**, covering scrapers, REST endpoints, and export pipelines.

Samyukt — Cloud Version Control Platform · *React 19, Node.js, Express, MongoDB, WebSockets* *GitHub*

- Architected a cloud-native version control platform featuring a **3-layer semantic merge engine** (text-level diff, AST structural parsing, and AI behavioral analysis) to automate code conflict resolution.
- Engineered database schemas and indexing across **6 core entities** in MongoDB Atlas / Mongoose with connection pooling and automated exponential backoff reconnection maintaining sub-30ms ping latency.
- Integrated real-time collaborative pipelines using **WebSockets & CRDT concepts**, enabling multi-user live presence tracking, cursor synchronization, and conflict-free concurrent editing.
- Containerized frontend/backend with **Docker & Nginx reverse proxy**, implementing automated test suites via **15+ Supertest integration specs** and **10+ Playwright E2E flows** with a 100% pass rate.

PresenceIQ — AI Attendance System · *Django REST, React, MongoDB, OpenCV* *GitHub*

- Built an automated AI facial recognition attendance platform with **Django REST**, **React**, **OpenCV**, and **InsightFace**, implementing role-based access control (RBAC) across 4 user tiers.
- Architected **7 MongoDB collections** for users, enrollments, sessions, biometric face data, attendance logs, and audit trails to enable scalable document management.
- Developed modular REST APIs across 3 core services for JWT authentication, class management, and automated facial recognition attendance logging.
- Integrated **PDF/Excel reporting**, attendance analytics visualizations, and bulk roster ingestion, validating the system with 20 student test profiles across live sessions.

EDUCATION

Bachelor of Engineering in Artificial Intelligence & Machine Learning 2023 — 2027
Visvesvaraya Technological University · CGPA: 8.14 / 10 *Bangalore, Karnataka, India*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Object-Oriented Programming

Higher Secondary Education (Class XII) 2022 — 2023
The Kingdom College · Percentage: 94.14% *Bangalore, Karnataka, India*

LANGUAGES

Spoken Languages: English (Professional), Hindi (Professional), Gujarati (Professional), Kannada (Professional)